Recommender Systems

- Large Language Models for Recommender Systems -

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LLMs in Recommendation[1,2,3]

[1]Zhao, Z., Fan, W., Li, J., Liu, Y., Mei, X., Wang, Y., Wen, Z., Wang, Z., Zhao, X., Tang, J., Li, Q. 2024. Recommender Systems in the Era of Large Language Models (LLMs). IEEE Trans. on Knowl. and Data Eng. 36, 11 (Nov. 2024), 6889–6907. https://doi.org/10.1109/TKDE.2024.3392335

[2]Hua, W., Li, L., Xu, S., Chen, L., Zhang, Y.. 2023. Tutorial on Large Language Models for Recommendation. In Proceedings of the 17th ACM Conference on Recommender Systems (RecSys '23). Association for Computing Machinery, New York, NY, USA, 1281–1283. https://doi.org/10.1145/3604915.3609494

[3]Giobergia, F. (2024) Lecture on Large Language Models. https://dbdmg.polito.it/dbdmg_web/2024/large-language-models-2024-25/

Language Models

- A **language model** is a probabilistic model for natural language
  - Prediction of the probability for any sequence of words $P(w_1, w_2, \ldots, w_t)$, e.g.:
    - $P(this, is, a, reasonable, sentence) = 0.1$
    - $P(this, is, a, purple, sentence) = 0.01$
    - $P(this, are, a, reasonable, sentence) = 0.001$
  - Plausible sentences have higher probabilities
    - E.g., grammatical correctness and coherent sequences of words
  - For language generation, the aim is to estimate the probability that the next word w_t, will follow other words
    - $P(w_t | w_{t-1}, \ldots, w_2, w_1) = ?$
  - Rather simple use cases, e.g., autocomplete or classification

Large Language Models

- A **large language model (LLM)** is a large version of a language model
  - Trained on massive datasets
  - Billions of model parameters
  - Complex natural language use cases
    e.g., summarization or reasoning

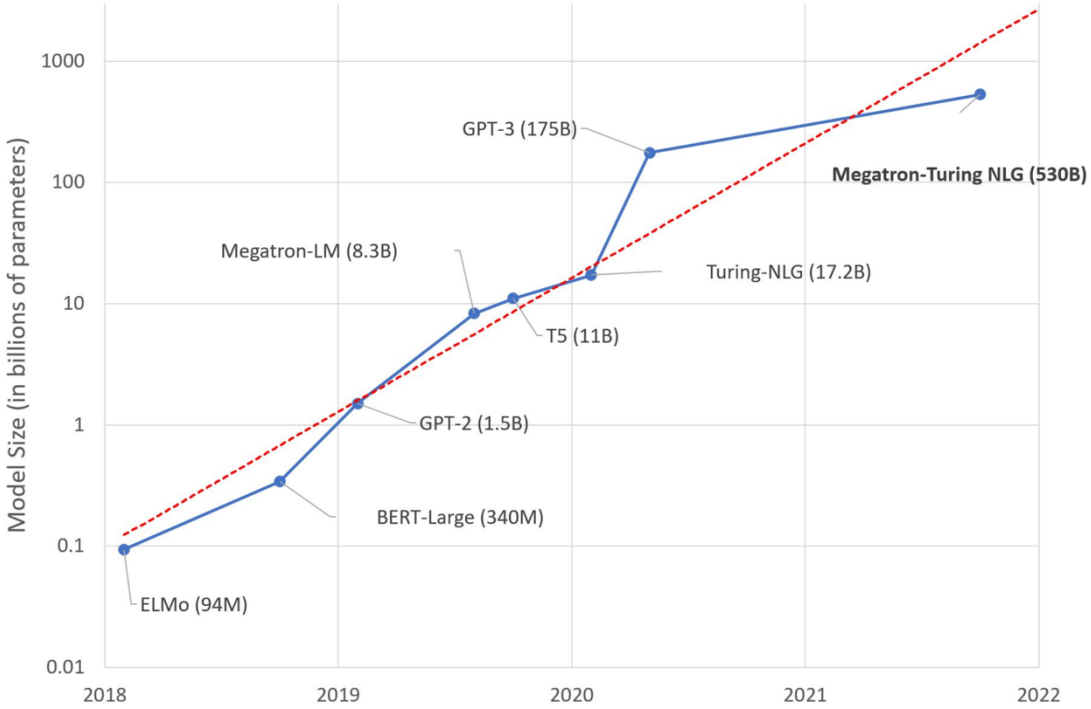
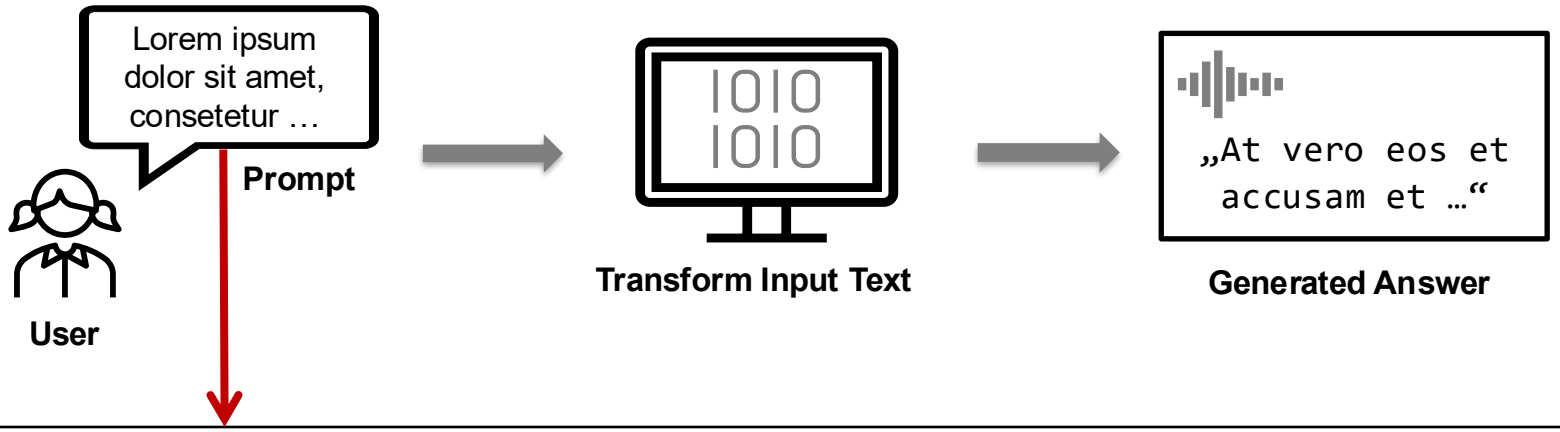


Image Source: https://huggingface.co/blog/large-language-models

Prompting

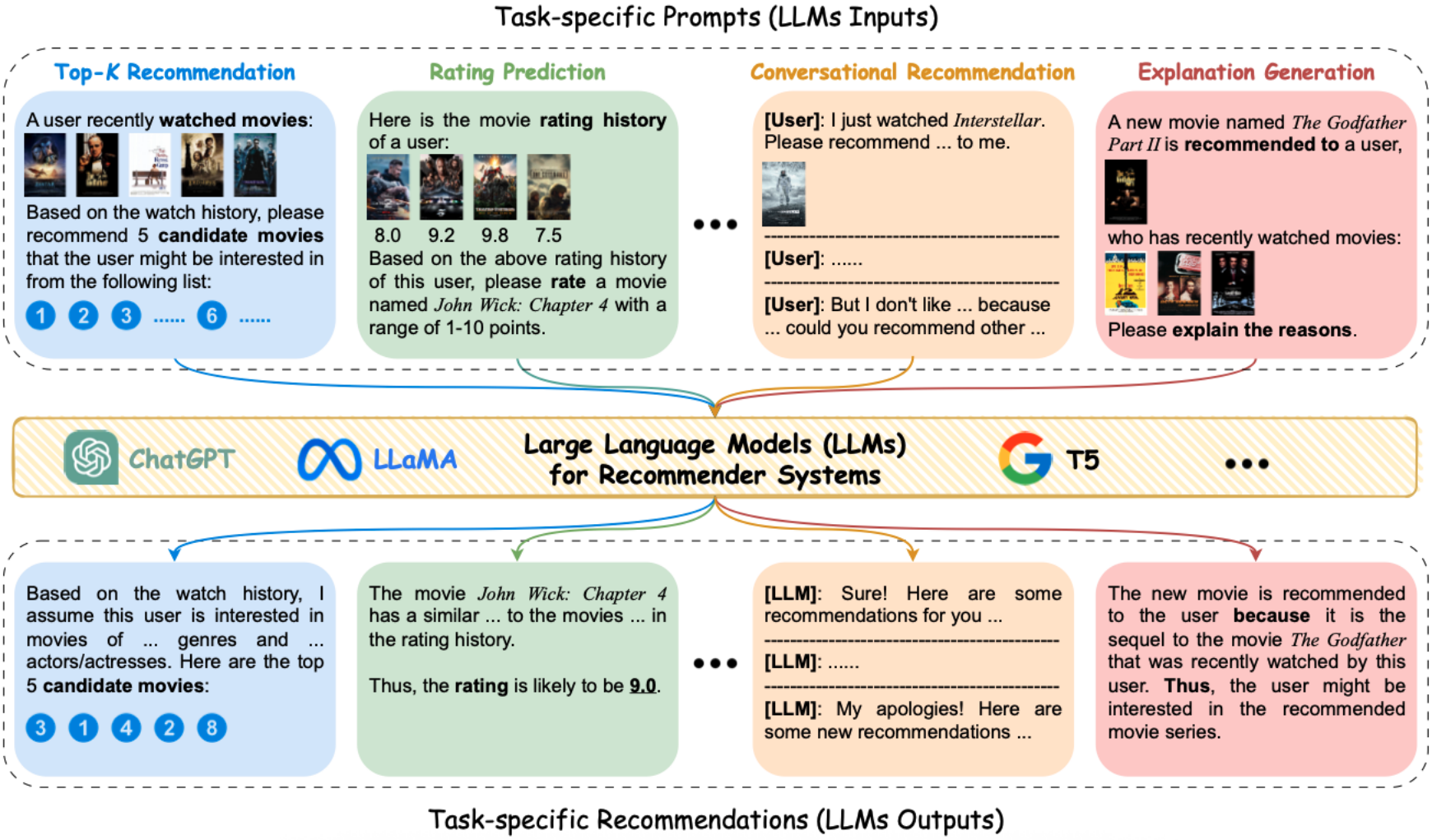
- LLMs are instructed with prompts




```
# System instructions (general behavior)
You are a chatbot answering user questions.
If the user asks for something illegal, explain that you cannot respond
        to this question.
Answer like a pirate.
[…]

# User instructions (explicit task description)
What is the most beautiful city in Austria?
```

Instructing LLMs for Recommendation Tasks



Motivation for LLMs in Recommendation

- Processing of unstructured data
  - Traditional recommenders require structured data
- Interpretation of the recommendation
  - LLMs can explain their reasoning (transparency)
- React to Feedback
  - LLMs can adapt recommendations with additional input
- Generalize from a few examples (zero-/few-shot learning)
  - Many traditional recommenders require much data for training

Excursus: Recommending Videos[1]

[1]Lubos S, Felfernig A and Tautschnig M (2023) An overview of video recommender systems: state-of-the-art and research issues. *Front. Big Data* 6:1281614. doi: 10.3389/fdata.2023.1281614

Video Recommender Systems

- A recommender system that recommends **videos** based on user preferences
- Different domains and application contexts

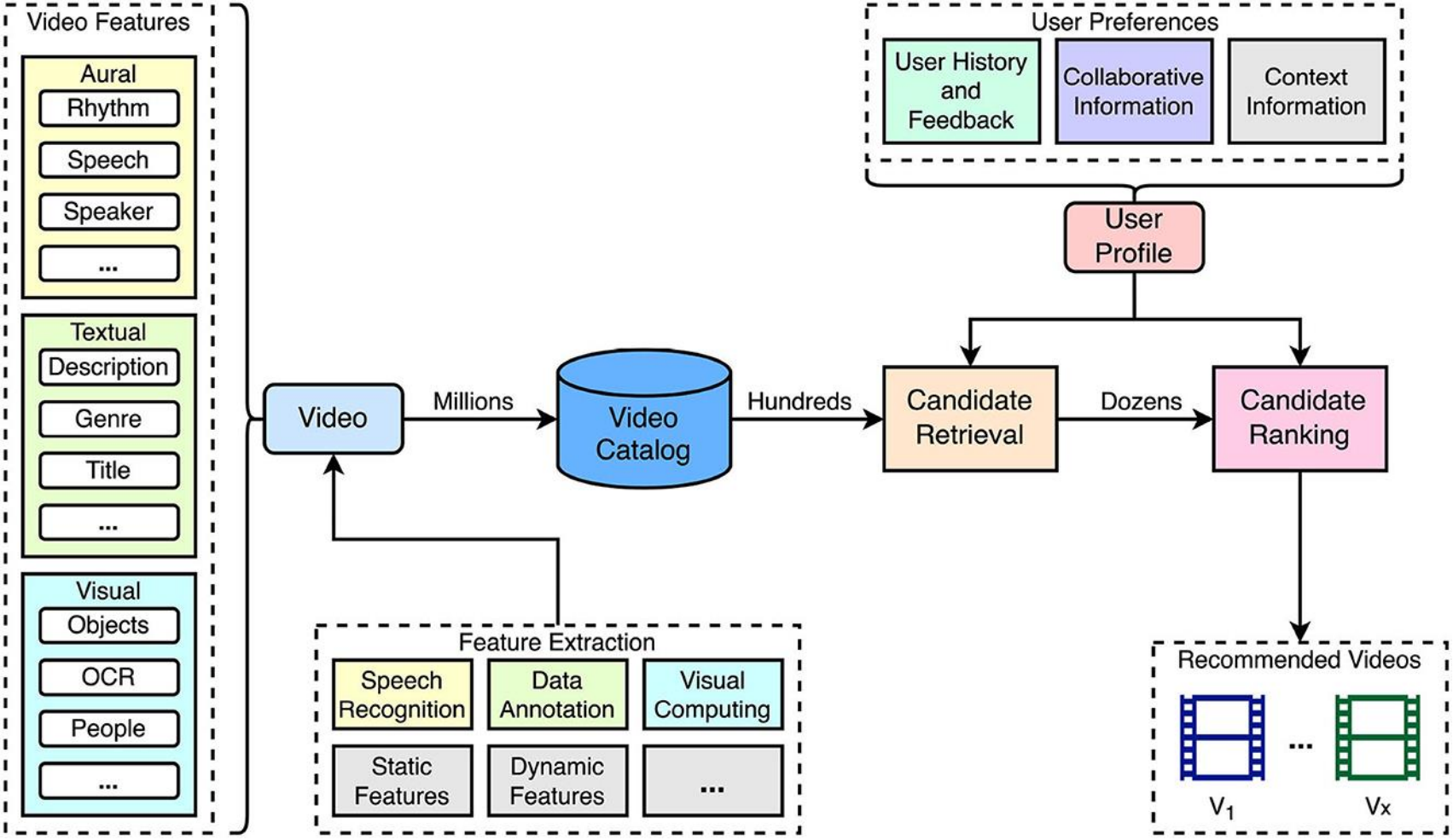
Domain	Application Context
Entertainment	• Movies and TV • Social Networks • Sports
Information	• News • Public political debates
E-learning	• Online courses • Skill building

→ High variability requires specialized concepts

Specific Challenges of Video Recommenders

- Multimodal content features
  - Audio, e.g., mood, rhythm, speech, speakers
  - Visual, e.g., colors, objects, people, written text
  - Textual, e.g., description, genre, title, transcript
- Influence of temporal dynamics, i.e., preference evolution
  - Trends, e.g., viral videos
  - Seasonal aspects, e.g., Christmas movies
- Dependency on user context, i.e., situation and environment
  - Time of the day, e.g., news in the morning and movies in the evening
  - Group of persons, e.g., horror movies when alone, and animation with kids
- Interpretation of engagement data
  - Meaning of interaction behavior and watch time
- Scalability for the massive volume of new items
  - Social media, e.g., TikTok, YouTube

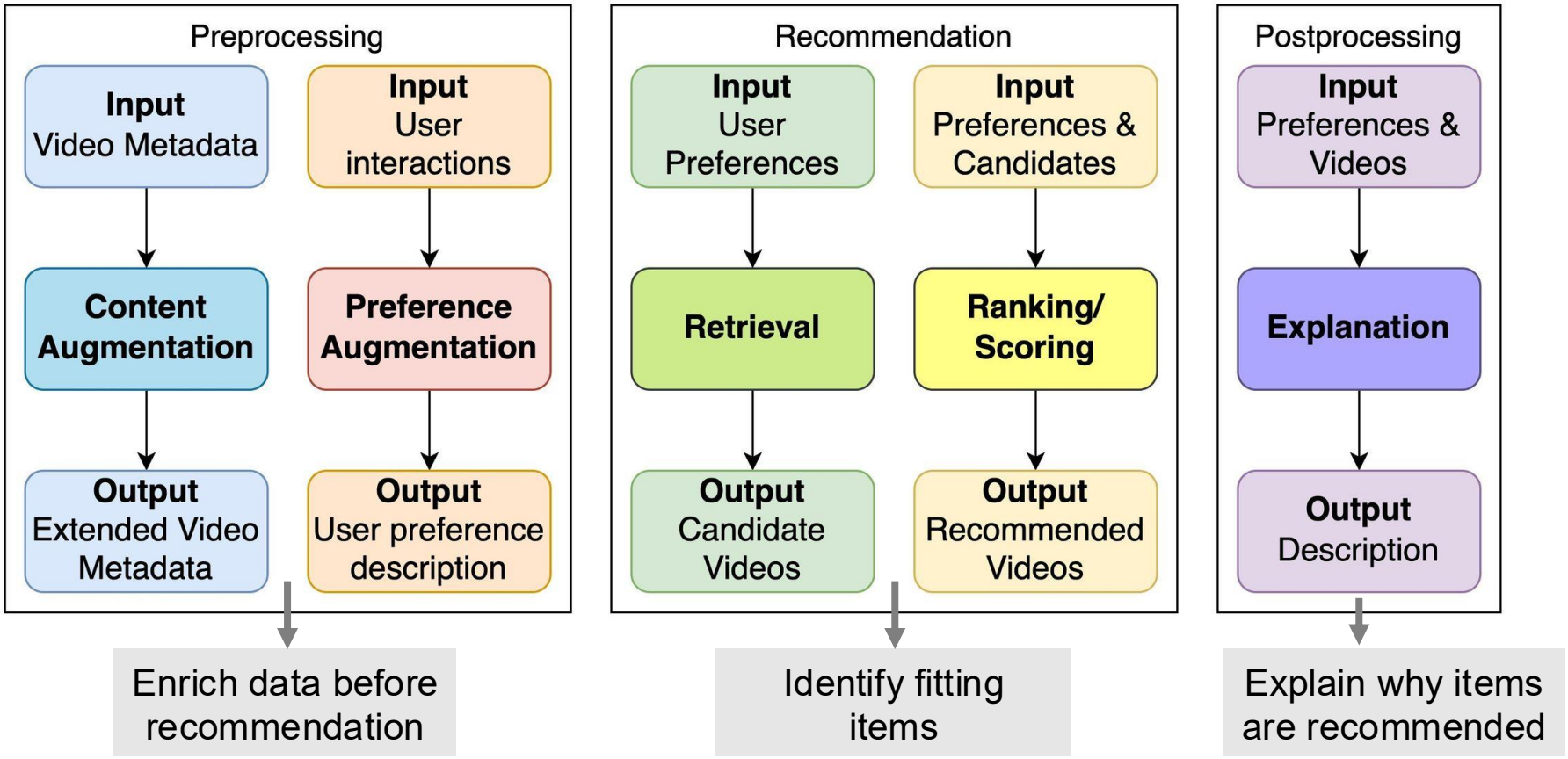
Video Recommender Pipeline



LLMs in Recommendation Tasks

LLMs in Recommendation Tasks

- Example for video recommender systems



LLMs for Content Augmentation

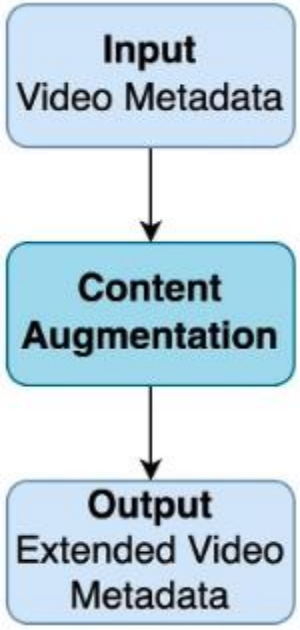
- Extend and enhance raw video metadata, e.g.:
  - Tag generation, summaries, extended genre labels
- Accessible and flexible option for metadata extension

Example – Video Classification:

```
Classify a recorded meeting based on its transcript.

You have two options:
– “sales meeting”: a meeting in which salespeople
  try to sell their product to customers
– “non-sales meeting”: a meeting with an arbitrary
  topic

Transcript: […]
```



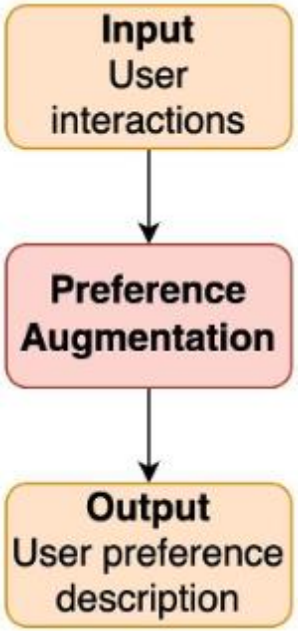
LLMs for Preference Augmentation

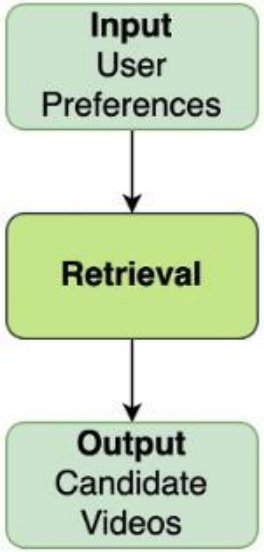
- Extend and enhance user profiles from interaction data, e.g.:
  - Analyze watched videos, compare watched videos, intent prediction
- Interpret the interaction data

Example – Intent prediction:

```
Based on the following video transcript, what was
the user's likely intent in watching this video?
Consider the following options:
– “learn”
– “solve a problem”
– “be entertained”
– “other”

Transcript: […]
```

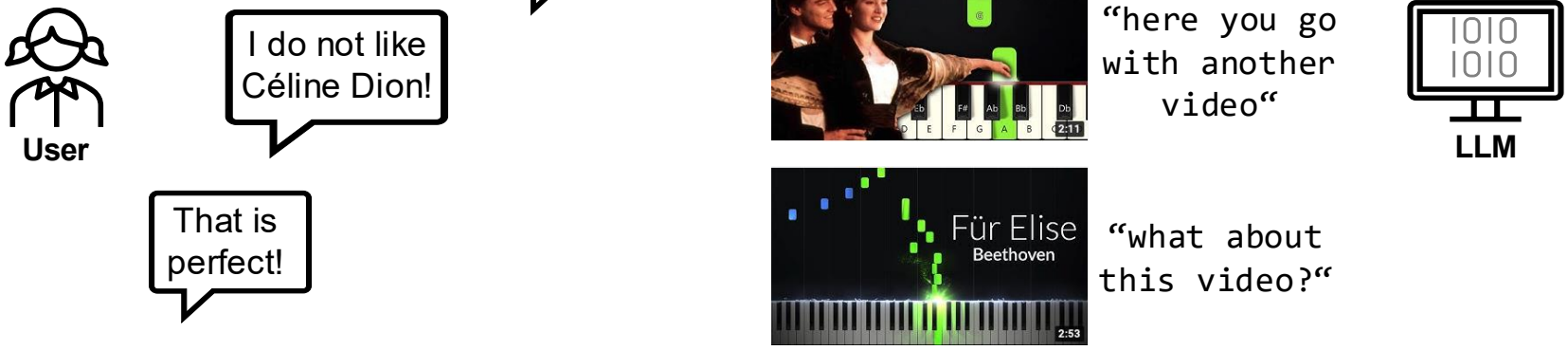




LLMs for Item Retrieval

- Identify relevant candidate videos based on the user preferences
  - Retrieve semantically relevant videos

Example – Critiquing-based Video Retrieval:



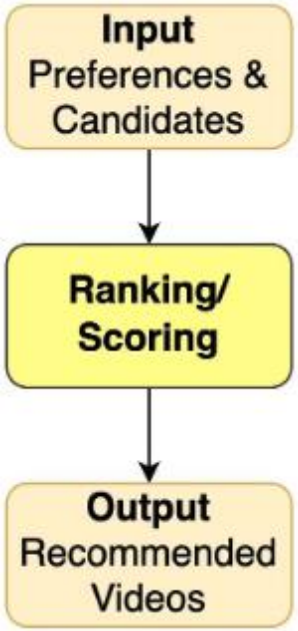
LLMs for Item Ranking

- Order candidate videos based on relevance
  - Consider user preferences to identify the best-fitting videos
- Interpret user preferences

Example – Intent-based Ranking:

```
Rank the following videos based on which fits
the user intent best.
User intent: Learn tennis serve

Video 1: "Top 5 Tennis Serves in History"
Video 2: "Step-by-Step Guide to Mastering Your
          Tennis Serve"
Video 3: "Funny Tennis Serve Fails Compilation”
[…]
```



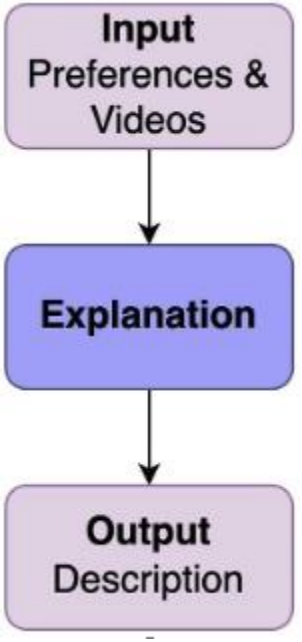
LLMs for Explanation of Item Recommendations

- Justify recommendations in natural language
  - Support users in determining whether a video fulfills their needs
- Interpret how user preferences and recommendations are connected

Example – Video Explanation:

```
Explain why the movie “Toy Story 3” was recommended
to the user.

Consider the last five movies watched by the user:
– Toy Story
– Minions
– Zoomania
– Toy Story 2
– Stuart Little
```

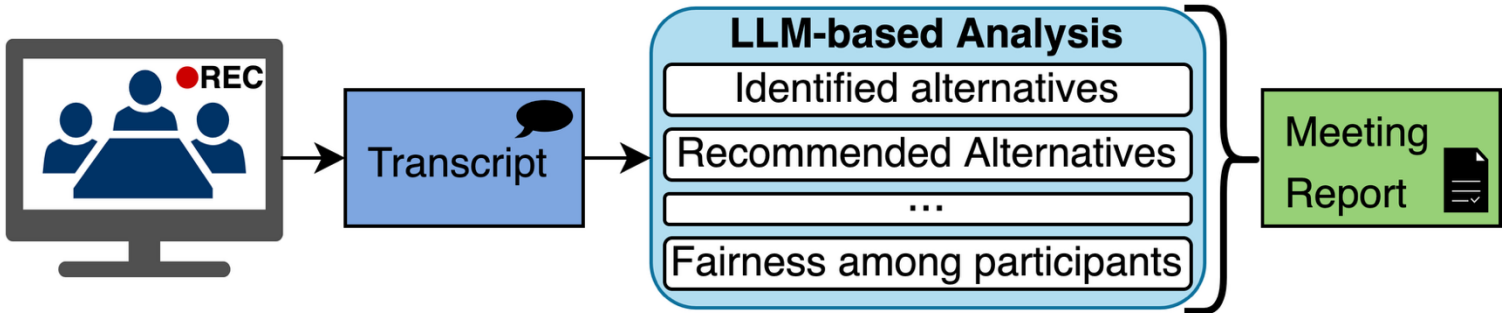


Item Generation

- LLMs can be used to generate relevant items
  - Recommendations beyond a fixed item catalog
  - Improve personalization
- Many use cases, e.g.,
  - Recommend titles, descriptions, and tags for uploaded videos
  - Recommend quiz questions for online lectures
  - Recommend further options based on group discussion
  - Recommend improvements for software requirement specifications
  - Recommend user stories for software feature descriptions
  - Recommend usability improvements for user interfaces
  - …

→ Recommendation of items that are not fixed in advance

Recommending Meeting Discussion Option[1]



- Derive insights and provide recommendations for recorded meetings based on the transcript
- Different usage scenarios, for example:
  - Task prioritization in SCRUM meetings
  - Product name decision
  - Alignment of project requirements
  - Destination planning for a group vacation
  - …

[1]S. Lubos et al., Towards Group Decision Support with LLM-based Meeting Analysis. Available: https://doi.org/10.1145/3708319.3733646

Requirement Specification Improvement[1]

LLM-based requirement quality evaluation (based on ISO 29148) and improvement recommendation.

Characteristic	Definition
Verifiability	The requirement is formulated in a way that its fulfillment can be proven or, in the best case, measured.
…	…

```
# Original Requirement
```
The DigitalHome programmable humidistat shall allow a user to monitor and control a home's humidity from any location, using a web ready computer, cell phone, or PDA.

LLM-based Evaluation

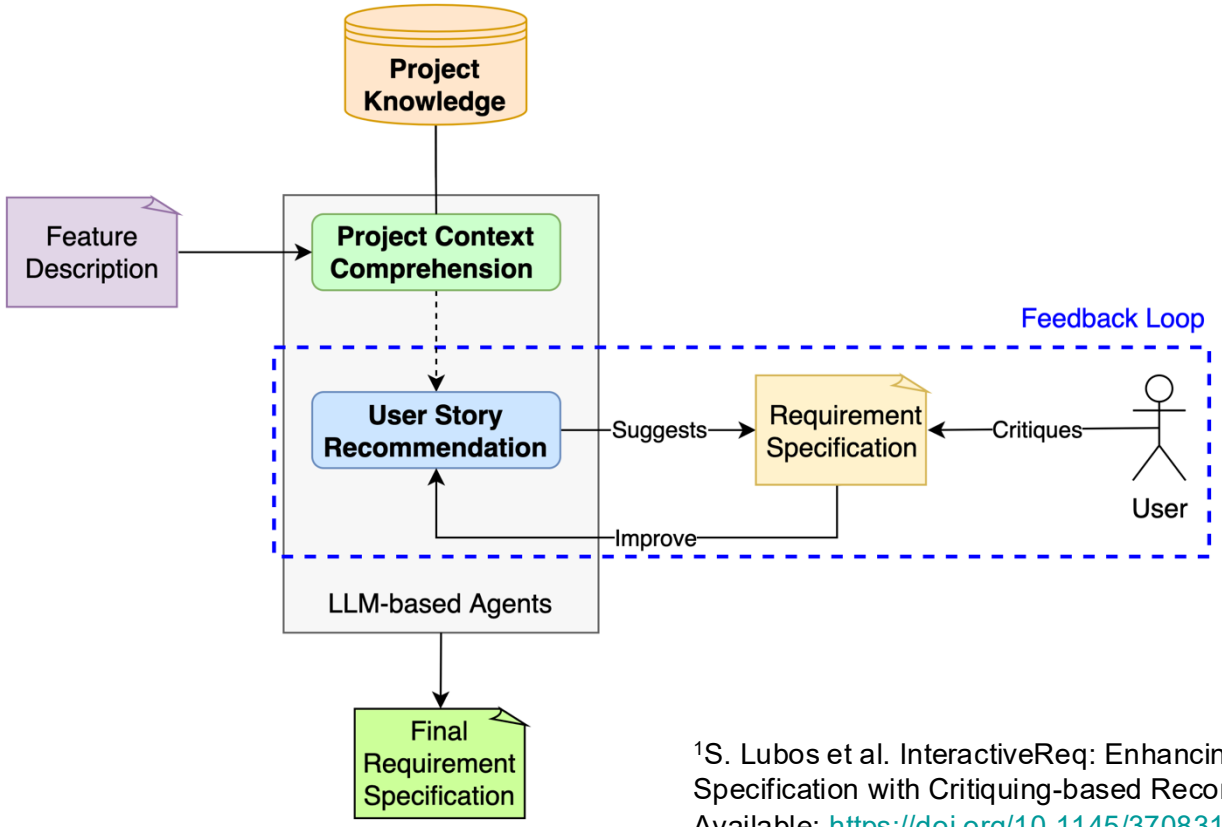
Decision: *not verifiable*
Explanation: *The requirement does not provide specific criteria for monitoring and controlling humidity. Additionally, the requirement does not specify how the user will be able to access the system remotely.*

Recommended Improvement: *The DigitalHome programmable humidistat shall allow users to monitor and control a home's humidity from any location, using a secure web-based interface accessible via a desktop computer, laptop, tablet, or smartphone. The system shall display real-time humidity levels and allow users to adjust the humidity level within a predetermined range, with notifications sent to the user's registered devices upon successful completion of the task.*

[1]S. Lubos et al., Leveraging LLMs for the Quality Assurance of Software Requirements. Available: https://doi.org/10.1109/RE59067.2024.00046

Interactive Requirement Specification[1]

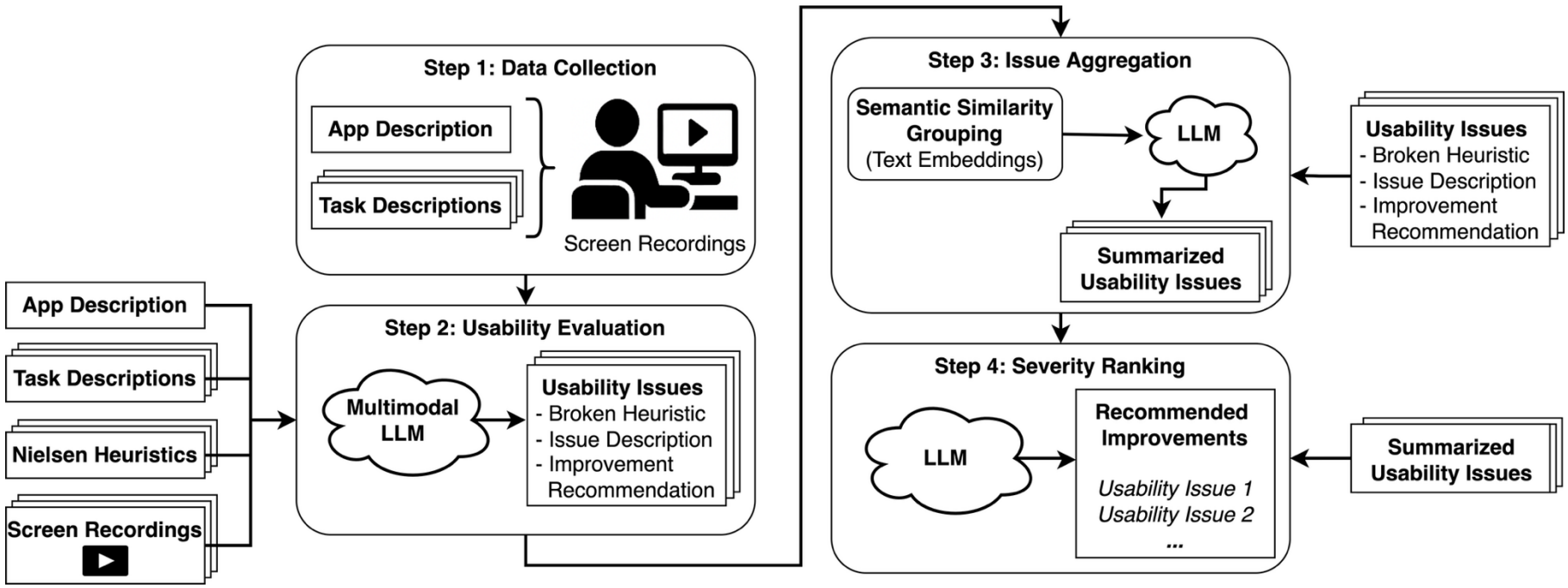
Specifying user stories through collaboration between stakeholders and LLM agents.



[1]S. Lubos et al. InteractiveReq: Enhancing Software Requirement Specification with Critiquing-based Recommender Systems. Available: https://doi.org/10.1145/3708319.3733678

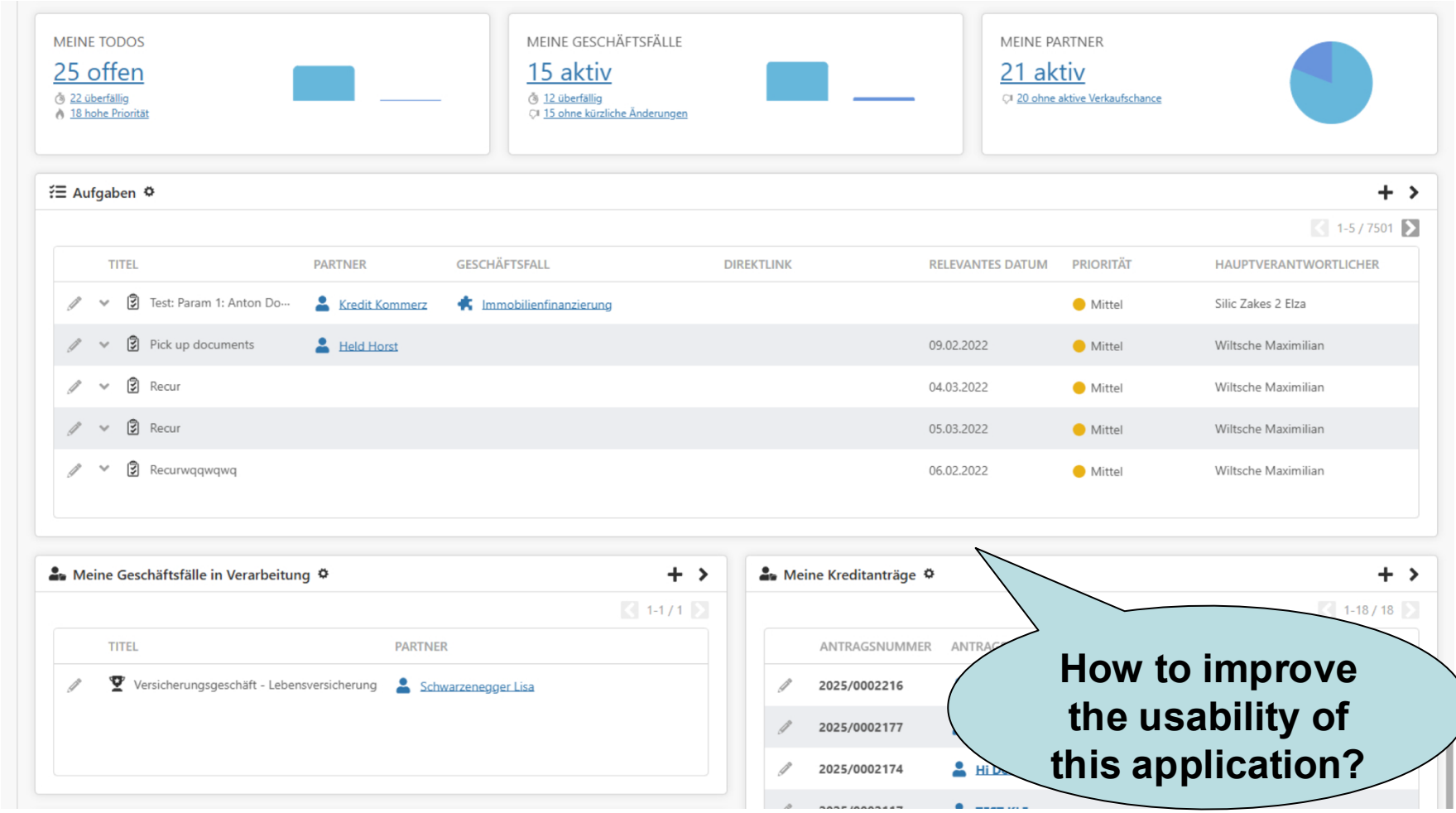
Usability Improvement Recommendation[1]

Use a multimodal LLM to evaluate the usability of user interfaces and recommend improvements.



[1]S. Lubos et al.. Recommending Usability Improvements with Multimodal Large Language Models. Available: https://doi.org/10.48550/arXiv.2605.29456

Usability Improvement Recommendation - Example



Usability Improvement Recommendation - Example

Usability Issue Description

The small bar and pie charts in summary sections like 'Meine ToDos', 'Meine Geschäfte', and 'Meine Partner' are very compact and lack immediate labels or legends within their visible area. This limited visual information reduces their immediate utility for a new user.



Usability Improvement Recommendation - Example

Usability Improvement Recommendation
Consider slightly increasing the size of these summary charts or adding minimal, high-level labels directly on or immediately adjacent to the charts. This would provide more context and improve immediate comprehension without requiring additional interaction.



Challenges

- LLMs are computationally expensive
  - Considerations for real-time performance are needed
  - Cost can be high
- Hallucinating LLMs
  - Generated output can sound plausible but may be false
  - ”Non-existent” items could be recommended
- Biased LLMs
  - Inherent bias from training data
  - Fairness is hard to control, e.g., popularity bias
- Complex evaluation
  - Performance can vary by prompt
  - Traditional metrics are often not applicable, e.g., generated items

Future Directions

- Multi-modal Recommendation
  - Use audio/video content as input for recommendations
  - E.g., video transcripts and encoded frames
- Continuous personalization
  - Adapt to user behavior over time to learn their preferences
  - E.g., using dynamic memory
- Agent-based Reasoning
  - LLM-based agent as a recommendation assistant
  - E.g., explaining, asking clarifying questions, planning

Thank You!